

Advanced (Habanero)

- Q1.** What is the mean of 2, 4, 6, 8, 10?
(A) 4
(B) 5
(C) 6
(D) 8
(E) 10
- Q2.** A data set has a median of 7 and a mode of 5. What can be said about the data set?
(A) It is symmetric
(B) It is skewed to the left
(C) It is skewed to the right
(D) The mean is 7
(E) The mean is 5
- Q3.** What is the mode of 1, 2, 2, 3, 4, 4, 4?
(A) 1
(B) 2
(C) 3
(D) 4
(E) No mode
- Q4.** A set of exam scores has a mean of 80 and a median of 75. What can be said about the scores?
(A) They are normally distributed
(B) They are skewed to the left
(C) They are skewed to the right
(D) The mode is 80
(E) The mode is 75
- Q5.** What is the median of 1, 3, 5, 7, 9?
(A) 3
(B) 5
(C) 7
(D) 9
(E) No median
- Q6.** Two data sets have the same mean but different medians. What can be said about the data sets?
(A) They are identical
(B) They are symmetric
(C) They are skewed in opposite directions
(D) One is skewed to the left and the other to the right
(E) No conclusion can be made
- Q7.** A data set has a mean of 10 and a standard deviation of 2. What is the z-score of 12?
(A) 1
(B) 2
(C) 3
(D) 4
(E) 5
- Q8.** Which measure of central tendency is most affected by outliers?
(A) Mean
(B) Median
(C) Mode
(D) Range
(E) Variance

Open Ended Questions (FRQ)

- Q9.** A company has two machines that produce light bulbs. Machine A produces bulbs with a mean lifespan of 1000 hours and a standard deviation of 50 hours. Machine B produces bulbs with a mean lifespan of 1050 hours and a standard deviation of 75 hours. Compare the two machines using measures of central tendency and variability.
- Q10.** A student scored 80, 70, 90, 85, 95 on five exams. Calculate the mean, median, and mode of the scores. Interpret the results in context.

Chef's Tips

- Remind students to show their work for FRQs
- Emphasize the importance of using correct units
- Make sure students understand the context of the questions